

Part 3 — Machine Learning Report

Smart City Traffic Intelligence — Capstone Project

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1. Experimental Setup

All experiments use the cleaned, feature-engineered dataset from Part 2 (data/processed/traffic_features.csv: 40,575 hourly records, 27 features). Models are evaluated with a **leakage-free time-based split**: training on all hours before 2018-01-01 (34,042 h) and testing on the whole of 2018 (6,533 h). Shuffling was deliberately avoided — for time series, a random split would let the model memorise neighbouring hours and overstate its skill.

Three prediction tasks were defined: **regression** (next-hour traffic volume), **congestion classification** (volume $\geq$ the Q3 threshold of 4,952 vehicles/h, 25.02% positive), and **high-risk screening** (congestion combined with severe weather or heavy rain > 5 mm — 594 positives, 1.46%).

2. Supervised Learning

Table 1 — Regression results on the 2018 test set (traffic_volume, vehicles/hour)

Model	MAE	RMSE	R ²
Naive baseline (predict training mean)	1,728.3	—	0.000
Linear Regression	652.3	854.1	0.8129
RandomForest (100 trees)	241.2	406.8	0.9575

Table 2 — Congestion classification on the 2018 test set

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression (scaled)	0.9218	0.8353	0.8563	0.8457	0.9626
RandomForest (100 trees)	0.9414	0.8695	0.9009	0.8850	0.9828

The RandomForest classifier's confusion matrix [[4677, 221], [162, 1473]] shows errors concentrated at the Q3 decision boundary — hours whose volume sits just either side of 4,952. Feature importances are dominated by calendar signals: hour_cos (0.185), is_rush_hour (0.149), hour (0.141), temp_c (0.108), dow_sin (0.073).

For the **high-risk screening** task (1.46% positives), the RandomForest with class_weight="balanced" reaches recall **0.949** at precision 0.689 (confusion [[6393, 42], [5, 93]] — only 5 of 98 high-risk hours missed in 2018). Recall was prioritised deliberately: for a screening aid, a missed high-risk hour costs more than a false alarm.

3. Unsupervised Learning

K-means clustering (13 scaled features, k chosen by silhouette score over $k = 2..5$) selected $k = 3$ (silhouette 0.1629, ahead of $k = 2$ at 0.1610):

Table 3 — Cluster profiles ($k = 3$)

Cluster	Hours	Avg volume	Congested	Weekend	Avg temp	Interpretation
0	11,596	2,624	3.1%	100%	8.3 °C	Weekend pattern
1	15,157	3,666	35.9%	0%	18.6 °C	Warm-season weekdays
2	13,822	3,438	31.5%	0%	−3.2 °C	Cold-season weekdays

Without any labels, clustering rediscovered the calendar: one pure-weekend cluster and two weekday clusters separated by season — corroborating the Part 1 finding that traffic is calendar-driven, with weather as a secondary modifier.

Association-rule mining (Apriori, min support 0.05, min confidence 0.5; itemset $40,575 \times 19$) found the strongest rules implying *High traffic*:

Table 4 — Top-5 association rules for "High traffic" (by lift)

Antecedent	Support	Confidence	Lift
{Afternoon 12–16, Clouds, Weekday}	0.052	0.741	2.96
{Afternoon 12–16, Weekday}	0.106	0.716	2.86
{Morning 06–11, Weekday}	0.105	0.593	2.37
{Afternoon 12–16, Clouds}	0.055	0.564	2.25
{Afternoon 12–16}	0.113	0.545	2.18

4. Advanced Analysis — LSTM and SHAP

A **PyTorch LSTM** (24 h lookback, 4 input channels, 4,897 parameters) forecasts next-hour volume from raw sequences. Evaluated on the same 2018 test set:

Table 5 — Next-hour forecasting comparison (2018 test set)

Model	MAE	RMSE	R ²
Persistence baseline (next hour = this hour)	588.9	814.0	0.8300
LSTM (24 h lookback)	389.4	539.4	0.9254
RandomForest (tabular features)	241.2	406.8	0.9575

The LSTM beats persistence clearly but loses to the RandomForest. The reason is instructive: the engineered calendar features (hour_sin/cos, is_rush_hour) hand the tabular model the daily rhythm directly, while the LSTM must re-learn it from raw sequences of limited length. Deep learning is not automatically superior — feature engineering is encoded domain knowledge.

SHAP (TreeExplainer, 1,000-hour sample of 2018) explains the congestion classifier. The top-5 features by mean |SHAP| — hour_cos (0.115), is_rush_hour (0.092), hour (0.064), dow_sin (0.037), is_weekend (0.037) — are all calendar features, matching both the Part 1 correlation analysis and the Day 6 importances. The beeswarm plot shows the direction: high is_rush_hour values push predictions toward congestion; weather features contribute small, situational adjustments. Per-prediction SHAP values allow an operator to audit any individual flag.

5. MLOps, Deployment and Monitoring

Experiment tracking (MLflow, SQLite backend): three runs (linear regression, RF regressor, RF classifier) logged with parameters, metrics and model artifacts; re-running the pipeline reproduces the reported metrics (RF MAE 241.2).

Deployment (FastAPI): a mock REST service exposes GET /health and POST /predict. The service performs feature engineering internally — callers send raw inputs (hour, day, weather) and the API builds the 27 training features, reading the column order from the model artifact itself (feature_names_in_) so training and serving can never drift apart. Five smoke tests pass: rush hour → ≈ 6,406 vehicles/h (HIGH), Sunday 03:00 → ≈ 450 (LOW), and invalid inputs are rejected with HTTP 422.

Monitoring (PSI drift + performance): the drift checker compares incoming batches against the training reference on the 15 environmental features (calendar features are deterministic and excluded; the reference window is Jan–Sep to match 2018's seasonal coverage). On real 2018 data it raised a **genuine single-feature ALERT**: the cloud-cover distribution shifted (PSI 0.267) while accuracy held (MAE 241.2) — so the documented action was *log and investigate, no retrain*. A synthetic heat-wave batch (+15 °C, 10× rain) tripped ALERT on exactly the right feature (temp_c, PSI 1.38), proving detection works. Retraining is triggered only when data drift coincides with performance degradation.

Recommendation system: a CLI advisor combines conditional congestion rates, the Day 6 rules and hourly profiles into advice for commuters and operations staff — e.g. weekday 17:00 Clear → HIGH (94.2% historical congestion, 3.77× base rate, shift to off-peak hours); Sunday 03:00 → LOW (maintenance window); weekday 08:00 Snow → HIGH (54.1%) with the matching rule {Morning, Weekday} → High traffic (lift 2.37).

6. Conclusion

Calendar structure dominates this corridor's traffic; weather modulates it at the margin. The strongest model is a RandomForest on engineered calendar features (R^2 0.9575); the LSTM and clustering provide independent confirmation of the same structure; and the MLOps layer (tracking, deployment, monitoring, recommendations) turns the models into an auditable, maintainable service. Limitations — proxy labels, single corridor, observed rather than forecast weather — are documented in the accompanying Responsible AI report.